DESIGN PRINCIPLES FOR THE CONSTRUCTION OF A BENCHMARK EVALUATING SECURITY OPERATION CAPABILITIES OF MULTI-AGENT AI SYSTEMS *

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ABSTRACT

***K*eywords** SOC-bench · Design principles · Multi-agent AI systems

1 Introduction

Security Operation Centers (SOCs) are addressing a fundamental societal challenge: enabling organizations (e.g., enterprises across various industry sectors, educational institutions, governments) to deal with cyber threats. The action-taking aspect of a SOC is supported by a sophisticated decision-making process, and this process involves a combination of both lower-order cognitive skills (e.g., summarizing, retrieval, recall) and higher-order cognitive skills (e.g., explanation, evaluation, reasoning). Accordingly, today's *cybersecurity operations* are heavily relying on manual effort, though security teams do leverage a variety of information technology tools such as Intrusion Detection systems, SIEM (Security Information Event Management) systems, and SOAR (Security Orchestration, Automation and Response) tool suites.

As Large Language Models (LLMs) and multi-agent AI systems are demonstrating increasing potential in cybersecurity operations, organizations, policymakers, model providers, and researchers in the AI and cybersecurity communities are interested in quantifying the capabilities of such AI systems to achieve more autonomous SOCs and reduce manual effort. In particular, the AI and cybersecurity communities have recently developed several benchmarks (e.g., Cybench [**?**], CyberGym [**?**]) for evaluating the *red team* capabilities (e.g., penetration testing) of multi-agent AI systems. Taking Cybench [**?**] as an example, it includes 40 professional-level Capture the Flag (CTF) tasks from 4 distinct CTF competitions, spanning a wide range of difficulties.

However, because the operations in a SOC are dominated by *blue team* operations, the capabilities of multi-agent AI systems to achieve more autonomous SOCs cannot be evaluated without a benchmark focused on blue team operations. (To the best of our knowledge, no systematic benchmark for evaluating blue team AI has been proposed in the literature.) The goal of this work is to develop a set of design principles for the construction of a benchmark, which is denoted as **SOC-bench**, to evaluate blue team operation capabilities of multi-agent AI systems.

Due to the fundamental differences between red team and blue team operations, the design principles we seek to develop are very different from the existing work focused on developing a benchmark evaluating read team AI. In order to validate these design principles, we have followed these principles and developed a conceptual design of SOC-bench, which consists of five XXX.

The rest of this paper is organized as follows. In Section 2, we explain why a benchmark focused on evaluating blue team AI agents would play an indispensable role in the technology adoption lifecycle of real-world SOCs in the era of AI. In Section 3, we present a set of design principles for SOC-bench. In Section 4, we provide an overview of the *conceptual design* of SOC-bench. From Section 5 to Section 9, we present the five major parts of our conceptual design. In Section 10, we conclude the paper with discussions.

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2 Why SOC-Bench

1. There are many existing benchmarks
2. Why there is still a need
3. SOC Landscape
   (a) The scope of SOCBench cannot be taken for guranteed
   (b) Why the midway on the spectrum is esential for development of agentic AI in cybersecurity

3 Design Principles of SOC-Bench

1. DP1: Loyalty to existing SOCs
2. DP2: Be exclusive
3. DP3: Based on real-world ransomware attack incident (try NOT to generalize)

4 Overview

4.1 Task Overview

Objective: SOC-Bench is a benchmark framework developed to evaluate AI agent capabilities across the full ransomware incident lifecycle in a SOC environment. The goal is to test not just detection and containment but also awareness, reasoning, and decision-making under realistic constraints. SOC-Bench does not create new tasks but replicates real SOC tasks and data from actual incidents. This approach ensures that the evaluation focuses on how AI agents would perform under the same operational conditions faced by real SOC analysts.

4.2 Terminology

- **SOC-Bench:** Benchmark framework that evaluates AI agents across the ransomware incident lifecycle in a SOC (Security Operations Center) environment.
- **Agent:** The AI system under evaluation that reasons from SOC-observable data and produces recommended actions or reports.
- **Agentic AI System:** An AI system composed of multiple coordinated agents that can reason autonomously and select actions across SOC tasks.
- **Task:** One of the five SOC-Bench problems (Fox, Tiger, Panda, Goat, Mouse) representing distinct SOC functions.
- **Telemetry:** Logs, alerts, network flows, host metrics, and other data collected by SOC systems.
- **Visibility Constraints:** Limits on the telemetry or data available to the agent during a task.
- **Ground Truth:** The reconstructed factual state of an incident observed from post-incident analysis (not available to agents during evaluation).
- **IOC (Indicator of Compromise):** A data point such as a hash, IP address, domain, or file path that signals potential malicious activity.
- **CTI (Cyber Threat Intelligence):** External or internal information about adversary behavior, tools, or indicators.
- **TTP (Tactics, Techniques, and Procedures):** The patterns, methods, or behaviors used by an attacker.
- **Inherent Inter-Task Dependencies:** Dependencies across SOC functions that create natural ambiguity, requiring agents to reason carefully to avoid incorrect conclusions.
- **Containment Action:** A recommended step, such as host quarantine, network segmentation, or process termination, intended to limit spread.
- **VSS Snapshot:** Windows Volume Shadow Copy snapshot used to preserve prior file states.
- **Encrypted File:** Windows Volume Shadow Copy snapshot used to preserve prior file states.
- **SIEM (Security Information and Event Management):** A system that collects and correlates security alerts, logs, and telemetry from across the network to support detection and incident analysis.
- **Lateral Movement:** The technique used by attackers to move from an initial foothold to additional systems within a network.

4.3 Scope

Blue Team Perspective: The benchmark reflects what a SOC sees during an enterprise ransomware incident. Inputs come from the defender side telemetry, logs, alerts, and employee reports. Attacker-side details or hindsight information are not included.

4.4 Network Topology

The SOC-Bench network mirrors the enterprise topology observed during the Colonial Pipeline ransomware incident. It mirrors the same segmentation weaknesses, host types, and traffic flow patterns that shaped the SOC's real response. Critical systems such as file servers, Active Directory controllers, VPN servers, and operational technology (OT) devices remain connected through limited segmentation, allowing partial lateral movement between subnets.

Each subnet represents a functional area of the enterprise, including corporate IT, OT, and remote access infrastructure. This structure allows AI agents to reason about containment and propagation within the same boundaries faced by human SOC analysts.

The topology is not redesigned or optimized but faithfully derived from post-incident data and telemetry to ensure authenticity.

This network context serves as the foundation for all SOC-Bench tasks. It allows evaluations to be conducted under realistic visibility constraints, ensuring that AI agents operate within the same physical and logical conditions as the original SOC investigation.

4.5 Ransomware Characteristics in Real-World Incidents

Attack Initiation: Most ransomware intrusions begin through a compromised access point, such as a VPN exploit. This aligns with Task Fox (Early Attack Detection), which evaluates whether agents can identify coordinated ransomware activity early but not prematurely based on real SOC data.

Lateral Movement: After initial access, attackers move laterally toward critical systems such as file servers and AD controllers. Because SOC-Bench reflects a network with limited segmentation, AI agents must reason about containment under realistic restrictions. This stage connects to Task Panda (Containment), which measures whether agents can detect spread and suggest isolation actions.

Impact: Once critical systems are reached, the ransomware encrypts files and disrupts operations. This phase is represented by Task Goat (File System Forensics), in which agents analyze metadata and logs to determine which files are encrypted, identify the responsible processes, and gauge the impact.

4.6 Threat Context and Data Complexity

Ransomware Basis: SOC-Bench tasks model ransomware behaviors observed in incidents such as Colonial Pipeline and DarkSide.

Incident Reconstruction: Multi-stage attack sequences are reconstructed from SOC-observed data, including lateral movement, privilege escalation, and decoy activity. Each reconstructed attack sequence reflects realistic SOC operations and is informed by incident telemetry and post-incident analysis. The benchmark is intentionally difficult, requiring agents to reason under uncertainty and partial information. This approach evaluates AI performance under similar operational conditions faced by real SOC analysts, rather than relying on artificial or synthetic environments. Scores are expected to be low at first, reflecting the complexity of genuine SOC workflows rather than deficiencies in model capability.

Data Integration: Data sources are unified, so agents must connect events across layers such as network, endpoint, and SIEM.

Detection Focus: Agents are rewarded for minimizing false positives and reasoning through uncertain evidence rather than matching static signatures.

4.7 Why SOC-Bench Can Focus on Five Representative Tasks?

- **Task Graph Flow:** SOC-Bench follows a five-task structure (Fox, Tiger, Panda, Goat, and Mouse) aligned with how a SOC processes ransomware cases from early detection to recovery.
- **Evaluation Priorities:** Scoring emphasizes analytic depth, situational awareness, reasoning quality, and evidence alignment over simple correctness.
- **Goal:** Encourage the development of AI systems capable of autonomous and explainable decision-making under incomplete information.

4.7.1 Task Fox – Early Campaign Detection

Task Fox evaluates the AI's ability to detect coordinated ransomware activity in its earliest observable stages using SOC-visible telemetry. The evaluation is stage-scored (e.g., 30-minute intervals), and at each stage, the agent must produce structured, evidence-based outcomes derived strictly from logs and telemetry, rather than inferred attacker intent. The evaluation occurs in three structured outcomes at every stage:

1. **First:** The agent must perform a Campaign-Scale Assessment, determining whether the observed activity is isolated, localized, or indicative of a campaign-scale event spanning multiple hosts. This determination must be supported by concrete cross-host telemetry evidence.
2. **Second:** If the activity is assessed as campaign-scale, the agent must perform Activity-Type Reasoning and determine whether the behavior reflects ransomware-like staging or coordinated non-ransom activity. This classification must cite at least two distinct supporting evidence objects and include a concise rationale.
3. **Third:** The agent must produce a SOC-style Incident Alert Triage Bundle that correlates cross-host signals and identifies the earliest stage where activity becomes campaign-scale, the earliest stage where it becomes ransomware-like, and the evidence-backed relationship between the two.

The task emphasizes structured, outcome-based reasoning under uncertainty, mirroring the observable signals available to Tier 1 SOC analysts and prioritizing precise evidence alignment over unsupported conclusions.

4.7.2 Task Goat - File System Forensics

Assesses whether agents can perform file-system forensics during active ransomware encryption and produce evidence-backed outcomes. Agents must:

1. Label files and directories as encrypted, not yet encrypted, or unknown using available metadata and logs, and cite evidence for each label
2. Aggregate results into host/share-level impact summaries, including counts and encrypted-byte fractions.
3. Detect backup/VSS snapshot deletion or suppression and report host, event, and timing with supporting evidence.
4. Attribute the primary encryptor process tree when process/EDR telemetry exists, reporting parent/child relationships, command lines, and bytes written where available.
5. Produce a one-page, structured summary of main findings suitable for executive review and cross-reference evidence IDs.

4.7.3 Task Panda - Containment

Task Panda evaluates whether AI agents can generate containment actions under uncertainty. The task includes natural ambiguity created by inter-task dependencies, so agents must differentiate between truly compromised assets and those that only appear suspicious without additional context. Agents must:

1. Detect when containment is required using multiple telemetry sources
2. Recommend a containment action and list the affected assets
3. Provide a concise rationale and cite supporting evidence
4. Avoid unsafe actions by reasoning through inter-task dependencies and recognizing when ambiguous evidence lacks sufficient context

The task focuses on reasoning, timeliness, and accuracy in containment decision-making, mirroring how SOC teams decide when and where to isolate assets during live ransomware attack campaigns.

4.7.4 Task Tiger – Threat Intelligence Analysis

Task Tiger evaluates the AI's ability to analyze IOCs, SIEM alerts, and CTI reports to identify attacker TTPs and accurately attribute the campaign, reflecting the real analytical workflow of Tier 2 SOC analysts. The goal is to determine whether agents can connect events, correlate across different data sources, and explain how the attack developed. Agents must:

1. Review IOCs, SIEM alerts, CTI reports, and other relevant telemetry
2. Identify the threat activity and attribute it to the correct actor or campaign
3. Describe attacker TTPs using evidence from the available data sources
4. Produce an intelligence summary that supports the later phases of the response
5. Prioritize accuracy and reasonable context in the analysis

The task evaluates whether the agent can reason like a Tier 2 SOC analyst during attribution, connecting what happened, why it happened, and how the attack unfolded.

4.7.5 Task Mouse – Specialized SOC Tasks (Data Exfiltration)

Task Mouse evaluates the AI's ability to detect whether data exfiltration occurred during the detected attack campaign and left the victim's environment. Using data such as firewall logs, network flows, and host events, agents must infer:

1. Whether exfiltration occurred
2. The approximate start time
3. The estimated range of exfiltrated data
4. Involved hosts where data exfiltration occurred
5. Any application-layer or network management protocols used

The task emphasizes fine-grained reasoning and evidence alignment under limited visibility, replicating real SOC conditions where exfiltration traces are subtle and incomplete.

5 Task Goat

5.1 Task Overview

(Principles: SOC-first, Outcome-only, Durability.)

Objective. Evaluate an autonomous SOC system on *file-system forensics* during a Colonial Pipeline–style enterprise ransomware incident in the business IT environment (Windows). The task checks the ability to: (i) label encryption at file and directory levels, (ii) roll up host/share impact, (iii) detect VSS tamper, (iv) attribute primary encryptor processes when telemetry exists, and (v) produce a one-page summary of main findings.

What must be submited. AI must produce **five outcomes** (details in Section 5.5):

1. O1: Encryption-state labels at file and directory levels.
2. O2: Host/share impact aggregations.
3. O3: VSS tamper facts (host, event, timing) with evidence.
4. O4: Attribution of primary encryptor process trees (when available).
5. O5: One-page *summary of main findings* referencing O1–O4 claims (*no new evidence in O5*).

Where inputs come from. Existing SOC data sources: file-system metadata and change journals (Sec. 5.4.1), process execution telemetry (Sec. 5.4.2), backup/VSS logs (Sec. 5.4.3), optional SIEM alert stream (Sec. 5.4.4), grader-facing reference file pairs (Sec. 5.4.5), and late help-desk reports written in employee phrasing (Sec. 5.4.6). Claims must reference resolvable `evidence_ids` in the pack's chain-of-custody.

Reader-only note. This task specifies *outcomes only*. Methods are deliberately unspecified; evaluation is by matching claims to ground truth via the scoreboard (Sec. 5.7).

5.1a. Design Principles (benchmarking)

- **SOC-first ordering:** The unit reflects what a SOC *observes and reports*, not the attacker's internal sequence.
- **Outcome-only:** Submissions are judged by claims/artifacts against ground truth; no subtasks or logic are mandated.
- **Hidden cross-task hints:** Inter-animal relations (GOAT, PANDA, FOX, TIGER, MOUSE) are *not* disclosed in task text to encourage emergent coordination.
- **Intentional incompleteness:** Some common signals (e.g., signature IDS rule-hits) may be withheld to avoid shortcutting via static signatures alone.
- **Durability:** Specification relies on stable OS/forensic constructs so it remains valid without changes for many years.

5.2 Terminology

(Principles: Durability; reused, plain terminology.)

- **Path:** absolute file or directory path (NTFS or SMB share).
- **Logical share:** Windows drive letter (e.g., `C:`) or UNC share (e.g., `\\server\share`).
- **VSS snapshot:** Windows Volume Shadow Copy snapshot.
- **Encrypted file:** content transformed by ransomware.
- **Ransom note:** small textual file deployed by the actor indicating extortion terms.

5.3 Scope and Assumptions

(Principles: SOC-first, Durability.)

- **Colonial-only focus:** incident dynamics mirror the **Colonial Pipeline** case (2021) [1, 2, 3].
- IT/business network only; no OT/ICS encryption is in scope.
- Windows endpoints and file servers; NTFS and SMB shares prevalent.
- Input is existing data sources; the task evaluates outputs at designer-defined *checkpoints/stages* using the scoreboard (Section 5.7), not runtime performance.

5.4 Data Sources (with signals)

(Principles: SOC-first, Intentional incompleteness; reader-only signal notes are not for agents.)

5.4.1 File-system metadata and change journals

- **Metadata:** path, size, modified time, owner, extension; directory membership.
- **Change journal / audit:** create, rename, write events; rename cascades; per-process writer where available.
- **Signals (reader-only):** rapid per-process write bursts; mass renames; rare-extension spikes; directory-level dispersion of small notes; content entropy/compressibility rise after rewrites (when chunked content is available). See [4] for encryption-loop characteristics; see [1, 2] for ransom-note dispersion and widespread content impact.

> **Reader-only disclaimer for Subsection 4.1:** These signal bullets document real phenomena for human readers and must not be exposed to participating agents or included in prompts. Agents infer from raw data; graders/pack designers may consult these when constructing ground truth.

5.4.2 Process execution telemetry (EDR/XDR)

(Principles: SOC-first, Outcome-only.) Suspicious applications trigger file-system operations; this subsection follows those processes.

- **Process tree:** parent-child, command line, signer, image hash.
- **Write-event summaries:** per-process counts and magnitudes of write operations over rolling windows; selected module load context.

- **Signals (reader-only):** writer attribution to suspected encryptor lineage; outlier write profiles aligned with encryption loops [4].

Note on 4.1 vs 4.2. 4.1 records OS/file-system events per path; 4.2 records *process-centric* execution context and aggregated write activity. They are complementary.

Reader-only disclaimer for Subsection 4.2: Signals are for readers only and must not be loaded into agents.

5.4.3 Backup/VSS logs

- **VSS ops:** create/delete/disable events; backup service warnings/errors.
- **Signals (reader-only):** snapshot deletion/disable prior to mass encryption; restoration viability degradation [2, 1].

Reader-only disclaimer for Subsection 4.3: Presence/ordering of VSS operations is part of ground-truth construction and tolerances. Do not reveal specifics to agents.

5.4.4 SIEM alert stream (auxiliary)

- **Alerts:** high-level summaries derived from lower-level sources; corroboration only.

5.4.5 Reference file pairs (grader-only resource)

(Principles: Outcome-only, Durability.)

- **Content:** representative pre-incident cleartext files and matching post-incident encrypted counterparts included in the replay pack.
- **Checks (reader-only):** hash deltas; header/magic changes; entropy/compressibility shifts; directory-level encrypted-byte fractions aligned with the file/path inventory in Sec. 5.4.1. See [4].
- **Exposure:** available to graders and pack designers only; never surfaced to participating agents.

Reader-only disclaimer for Subsection 4.5: Exact file selections, hashes, and byte-level comparators establish ground truth and thresholds. Do not reveal filenames or checksums to agents.

5.4.6 Help-desk incident reports (employee phrasing)

(Principles: SOC-first; plain, non-technical descriptions.)

- **Content:** tickets or messages from employees describing symptoms such as "files will not open," "text looks scrambled," or "ransom note on desktop."
- **Signals (reader-only):** human-reported and typically later than telemetry; solid for narrative context and scoping, not for exact timing or byte-level state.
- **Scope:** not part of ground-truth computation for encryption labels; phrasing examples are illustrative and must not coach agents.

Reader-only disclaimer for Subsection 4.6: Do not call these reports "artifacts," and do not feed example text to agents.

5.5 Outcomes Overview

(Principles: Outcome-only; no agent advice.)

O1. Encryption-state labels. *Truth.* A file is *encrypted* if its bytes match the post-encryption reference (Sec. 5.4.5) and/or a decryptor/restore is required; a directory is *encrypted* if its encrypted-byte fraction exceeds the manifest threshold.
Submit. JSON list `encrypted_paths` with {`path`, `label`∈{encrypted, not-yet-encrypted, unknown}, `evidence_id`}.

O2. Host/share impact. *Truth.* Fraction of bytes encrypted and an impacted boolean per share/host derived from O1.
Submit. JSON `impact_by_host_share`: {`host`, `share`, `encrypted_bytes`, `total_bytes`, `first_seen`, `encrypted_bytes_fraction`}.

O3. VSS tamper. *Truth.* Snapshot delete/disable events prior to or during encryption waves on affected hosts.
Submit. JSON `vss_events`: {`host`, `event_type`, `time`, `evidence_id`}.

O4. Attribution (when available). *Truth.* Primary encryptor lineage is the process tree responsible for the majority of encrypted bytes per host.
Submit. JSON `attribution`: {`host`, `pid`, `parent_pid`, `cmdline`, `bytes_written`, `evidence_id`}.

O5. One-page summary of main findings. *Truth.* A single-page summary accurately reflects O1–O4 and introduces no new claims or evidence. It may cite `evidence_ids` that already appear in the O1–O4 JSON.
Submit. One-page PDF summarizing top impacts/findings and referencing O1–O4 (by `evidence_id` or claim identifiers); do not include raw evidence in O5.

5.6 Activity–Dependency Diagram

(Principles: SOC-first; observed order and dependencies only, no method implied.)

Reader-only disclaimer for Section 6: Diagram reflects *observed* ordering and dependencies in replay packs. It does not prescribe any method and should not be used to coach agents.

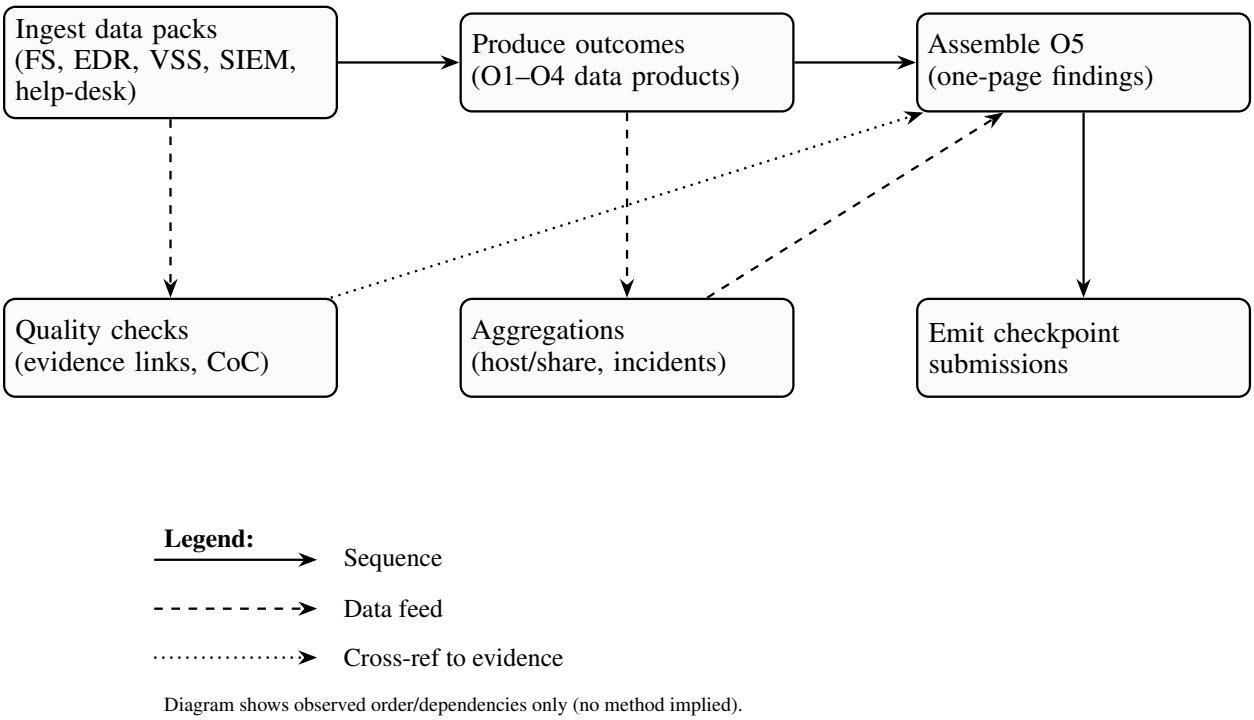

Figure 1: Observed activity–dependency for Task Goat.

5.7 Scoreboard

(Principles: SOC-first, Outcome-only.)

At each checkpoint, the manifest defines *targets* for O1–O5. Submissions provide *claims* with `evidence_ids` resolvable to pack sources. Scoring is ring-based per target; penalties are applied within the same flow so graders do not "finish then revisit." All numeric/time matching follows manifest tolerances known to graders (not to agents).

5.7.1 Rings and matching (applies to O1–O4)

For each target, consider only the best matching claim:

- **File-level exact:** correct file with required detail. *Exact*

- **Directory-level:** correct directory but not the specific file, or minor incompleteness at directory granularity. *Directory*
- **Host/share-level:** only the correct host/share is indicated. *Host/Share*
- **Miss:** incorrect or unsupported. *Miss*

Mistakes to avoid (grader). Do not map multiple claims to the same target; do not award any ring unless a valid `evidence_id` resolves in chain-of-custody; do not infer correctness from SIEM summaries alone.

5.7.2 General penalties (apply as you grade)

- **Wrong assertions:** each wrong claim that asserts a state (e.g., marks a path as encrypted when manifest disagrees) incurs -1 for that outcome, capped at 20% of the outcome's max at that checkpoint.
- **No evidence:** claims lacking a resolvable `evidence_id` are ineligible for ring points and incur -1 each (cap 10% of outcome max).
- **Contradictions:** mutually contradictory claims within the same stage about the same target yield -2 for that pair.
- **Stage leakage:** citing evidence that appears only at a later stage to justify a current-stage claim yields -2 per occurrence.
- **Duplicate spam:** duplicated claims beyond the first for the same target incur -0.5 each (cap 10% of outcome max).
- **Over-submission spam:** let B_X be the budget (max evaluated claims) for outcome X at a stage, and let C_X be total submitted claims for X. Define a spam threshold $S_X = 2B_X$. If $C_X > S_X$, apply

$$P_{\text{spam}}(X) \;=\; -\left\lfloor \frac{C_X - S_X}{B_X} \right\rfloor,$$

  i.e., -1 per full extra budget beyond $2\times$ over-submission.

5.7.3 Outcome-specific targets, points, and common grading mistakes

O1 Encryption-state labels (40 pts). Targets: 20 file/dir items. Ring mapping per item: *Exact* $= 2.0$, *Directory* $= 1.0$, *Host/Share* $= 0.5$, *Miss* $= 0$.
Common mistakes (penalize as in Sec. 5.7.2): trusting ransom-note presence alone; labeling a directory encrypted without meeting the manifest threshold; claiming *unknown* when truth is known.

O2 Host/share impact (25 pts). Targets: 5 impact facts with encrypted-byte fraction intervals. Per fact: *Exact* $= 5$ (correct scope and interval overlap), *Directory* $= 3$ (scope correct, interval off), *Host/Share* $= 1$, *Miss* $= 0$.
Common mistakes: aggregating across hosts; ignoring `first_seen`; intervals that do not overlap the manifest interval.

O3 VSS tamper (15 pts). Targets: 3 host-level facts $(\text{host}, \text{event}, \text{time})$. Per fact: *Exact* $= 5$, *Directory* $= 3$ (host correct, timing vaguely stated), *Host/Share* $= 1$ (only host impacted), *Miss* $= 0$.
Common mistakes: conflating generic backup errors with VSS snapshot delete/disable; timing after encryption waves.

O4 Attribution (10 pts). Targets: 2 host-level attribution facts (top writer process tree with parent/cmdline). Per fact: *Exact* $= 5$, *Directory* $= 3$ (lineage partially specified), *Host/Share* $= 1$, *Miss* $= 0$.
Common mistakes: attributing to a helper tool instead of the encryptor lineage; omitting parent or cmdline; relying on SIEM text without process evidence.

O5 One-page summary (10 pts). Checklist items (binary, 2 pts each): (i) covers O1 labels, (ii) covers O2 impact with numbers, (iii) covers O3 with host/time, (iv) covers O4 when telemetry exists, (v) references O1–O4 by claim/evidence identifiers without introducing new evidence in O5.
Common mistakes: marketing language with no references to submitted claims; numbers inconsistent with O1–O4 JSON; placing raw evidence or new claims in O5.

5.8 Compliance and Safety

(Principles: Durability, Outcome-only.) Read-only analysis; no actions against hosts. All claims must reference `evidence_id` entries resolvable in the pack. Reader-only subsections are for graders and must not coach participating systems.

6 Task Panda

6.1 Task Overview

Task Panda evaluates a multi-agent AI system's ability to reason over numerous data sources and generate timely and accurate containment responses during a Colonial Pipeline enterprise ransomware incident. In line with design principle 2 (DP2), this task focuses solely on containment actions, without considering other incident response phases.

Following design principle 3 (DP3), the scenario unfolds along a timeline divided into 30-minute stages. At each stage, the AI system receives a variety of data sources reflecting system activity, network traffic, performance metrics, and threat intelligence. These data sources are provided to simulate the data sources available to a real Security Operations Center (SOC), in accordance with design principle 1 (DP1). Detailed descriptions of these data sources are provided in the Data Sources section.

At each stage, the AI system must synthesize information from the data sources to determine whether and how to act effectively and produce a BLUF (Bottom Line Up Front) report summarizing the containment actions, affected assets, supporting evidence and reasoning. Reporting requirements and formatting are detailed in the Outcomes section.

For every stage, SOC Bench defines the ground truth representing the optimal SOC response. A scoreboard evaluates the AI system's accuracy, completeness and reasoning of containment actions based on the submitted BLUF reports.

Reader-only disclaimer for Task Overview: This task also introduces inherent inter-task dependencies to test the AI system's reasoning capabilities; however, such dependencies are not supposed to be disclosed to AI system developers or the AI systems.

6.2 Data Sources

All data sources provided to the AI system simulate the data sources available to a real SOC during the ransomware incident. In accordance with DP1 and DP3, the data strictly follows the Colonial Pipeline ransomware incident, reflecting the actual attack process and operational context. These sources are time-stamped and synchronized to provide realistic evidence for containment decision-making.

6.2.1 Filesystem Operations

Records of file modifications, creations, deletions, and encryption activity on monitored endpoints and storage systems. The corresponding logs provide insight into which files are impacted and the progression of ransomware activity.

6.2.2 Network Traffic

The corresponding PCAP files reveal communication patterns, lateral movement, command-and-control activity, and abnormal network behaviors.

6.2.3 Operating System Logs

Operating system logs provide visibility into authentication activity, process execution, service operations, and overall system behavior. Windows Event Logs capture security events such as logon attempts, process creation, PowerShell execution, and service or registry changes. Linux journal logs (journald) record system messages, authentication attempts, privilege escalations, service restarts, and kernel or daemon alerts. These logs help correlate host-level activity with ransomware behavior and support evidence-based containment decisions.

6.2.4 Host-level Measurements

Time-series measurements of CPU usage, memory consumption, I/O activity, and network throughput. Spikes or deviations can indicate ransomware operations or other anomalous behavior on hosts.

6.2.5 Cyber Threat Intelligence (CTI)

CTIs are Structured reports including indicators of compromise (IOCs), tactics, techniques, and procedures (TTPs) relevant to ransomware campaigns. CTI provides contextual knowledge for interpreting observed events and supporting containment decisions.

6.2.6 Network Topology Manifest

A structural mapping of hosts, VLANs, segments, and firewall zones in the simulated environment. This allows the AI system to understand containment boundaries and logical relationships between assets.

6.3 Outcomes

The AI system may produce a BLUF report at each stage; this report is the only evaluable output of Task Panda. It captures the agent's containment actions, affected assets, and the reasoning supporting these decisions. All expected outcomes follow the real-world Colonial Pipeline ransomware incident, in accordance with DP3.

If the agent determines that no containment action is necessary, it may still submit a BLUF report that omits actions and targets but provides reasoning and evidence explaining why inaction is justified. This mirrors real SOC decision-making, where inaction can be the correct and evidence-based choice when the available data does not support initiating containment.

6.3.1 BLUF Report Content Requirements

Every BLUF report must be structured in JSON format and must include the fields below. Each field is evaluated directly on the scoreboard.

(1) Stage timestamp

The end time of the stage when the decision is made.

(2) Containment actions

A list of all containment actions the agent proposes for this stage. Multiple actions may be included (e.g., isolations, firewall blocks, credential resets).

(3) Action targets

A structured list of all targets affected by the containment actions. Since real containment may involve many assets, this field supports multiple categories:

- hosts: endpoint hostnames or IDs
- ips: individual IP addresses
- subnets: network segments or VLANs
- user_accounts: individual compromised accounts
- user_groups: AD or IAM groups
- services: specific processes or enterprise services
- other: any additional SOC-relevant categories

Agents must list only assets directly affected by their containment actions.

(4) Reasoning and evidence

A concise BLUF-style justification that integrates three required elements:

(I) Action Justification Based on SOC Trade-Offs

The agent must explain why the selected containment actions are appropriate considering the trade-offs between:

- data security impact
- availability impact
- business/operational continuity

(II) assessment of Current Situation

The reasoning must assess the agent's evaluation of current situation based on data sources, such as:

- encryption activity
- lateral movement

- C2 connections
- exfiltration attempts
- compromised accounts
- anomalous host or network behavior

The assessment report must include concrete evidences (e.g. log records). This assessment must support the chosen actions and targets.

(III) Predicted Impact if No Action Is Taken

A brief forward-looking statement describing likely consequences of inaction, such as:

- ransomware spreading to adjacent hosts
- privilege escalation
- loss of business systems
- increased data exfiltration
- service outages
- degraded operational technology (OT) availability

This aligns BLUF reasoning with real SOC decision-making.

6.3.2 BLUF Report Format Requirements

All BLUF reports must follow standardized JSON formatting to ensure consistent scoring. Each report must be:

- structured (JSON only)
- self-contained
- time stamped
- explicitly listing actions, targets, and reasoning

Example JSON-formatted BLUF Report:

```
{
"stage_timestamp": "01/01/2025, 00:00:00 AM",
"containment_actions": [
"Isolate ENG-WKS-12 and ENG-WKS-17 from the corporate LAN",
"Block outbound SMB and HTTPS traffic from ENG-SRV-05 to 185.199.110.0/24"
],
"action_targets": {
"hosts": ["ENG-WKS-12", "ENG-WKS-17", "ENG-SRV-05"],
"ips": [],
"subnets": ["185.199.110.0/24"],
"user_accounts": [],
"user_groups": [],
"services": [],
"other": []
},
"reasoning_evidence": "Isolation and outbound blocking are selected to maximize security impact while minimizing operational disruption: both workstations are non-critical endpoints and blocking ENG-SRV-05 outbound traffic preserves service availability while halting lateral movement. Current data shows ENG-WKS-12 and ENG-WKS-17 exhibit rapid encryption and unauthorized C2 connections. ENG-SRV-05 is initiating anomalous SMB flows to a malicious subnet. If containment is delayed, ransomware will likely propagate across engineering hosts and enable further data exfiltration through ENG-SRV-05."
}
```

6.4 Scoreboard

The AI system's performance is evaluated per stage, based on the submitted BLUF reports. The scoreboard measures the accuracy, completeness, and quality of reasoning for containment decisions.

6.4.1 Scoreboard Components

(1) Containment actions—Maximum 20 points

Evaluates whether the containment actions match the corresponding stage's ground truth defined by SOC Bench.

Level	Points	Description
Fully correct	20	All actions exactly match the ground truth.
Partially correct	10	Actions address part of the ground truth but are incomplete or partially misapplied.
Incorrect or missing	-20	Actions are completely wrong, absent, or mostly incorrect, increasing operational risk and potential consequences.

(2) Action Targets—Maximum 40 points

Evaluates whether the agent correctly identifies all assets affected by the containment actions.

Level	Points	Description
Fully correct	40	Every affected asset is correctly identified.
Partially correct	20	Some assets are correctly identified, but key targets are missing or extra irrelevant assets included.
Incorrect or missing	-20	Most or all targets are incorrect or missing, which would likely cause ineffective containment or considerable collateral impact.

(3) Reasoning and Evidence—Maximum 40 points

Evaluates whether the agent provides coherent reasoning as described in Section 3.

Level	Points	Description
Fully supported	40	Reasoning is clear, logical, and fully supported by evidence from the provided data sources.
Partially supported	20	Reasoning is somewhat clear but incomplete or partially unsupported by evidence.
Unclear or unsupported	-20	Reasoning is unclear, illogical, or unsupported.

- "No-action" BLUF reports: If the agent submits a report without containment actions or targets, these fields are scored according to the ground truth (which may itself prescribe "no action" and "no target"), and Reasoning and Evidence are evaluated accordingly.

- "No-action" BLUF reports: If the agent submits a report without containment actions or targets, these fields are scored according to the ground truth (which may itself prescribe "no action" and "no target"), and Reasoning and Evidence are evaluated accordingly.

Reader-only disclaimer for Scoreboard: SOC Bench may embed inherent inter-task dependencies to test the AI system's reasoning capabilities. These dependencies are not supposed to be disclosed to the AI system or developers.

6.4.2 Score Calculation

Since the total number of stages is fixed, the Final Score is computed as the sum of per-stage scores:

Where:

StageScore_i = ContainmentDecision + ActionTargets + ReasoningAndEvidence + TrapPenalty

- N = total number of evaluation stages
- Range per stage: -80 to 100 (including trap penalties)
- Final Score range: -80 * N to 100 * N

6.5 Remarks

Task Panda evaluates AI systems per stage, with stages occurring every 30 minutes. Each stage has a ground truth used for scoring.

Effective containment requires correlating multiple independent data sources to identify valid actions.

Timeliness is inherently reflected in scoring: accurate and well-reasoned containment decisions submitted in a timely manner are rewarded. Premature actions without sufficient evidence or delayed responses that miss the stage window are implicitly penalized via negative scores.

All data sources are timestamped and synchronized to 30-minute stage to ensure consistent temporal alignment with real SOC operations, which is in keeping with DP1.

Reader-only disclaimer for Remarks: The inherent inter-task dependencies indicate that individual agents could suffer a lot if they do not coordinate with other agents; coordination is often required to avoid subtle misleading indicators. The dependencies are not supposed to be disclosed to the AI system or its developers.

6.6 Challenges

Reader-only disclaimer for Challenges: The content in this section is not supposed to be disclosed to AI system developers or AI systems.

Designing Task Panda involves several challenges that directly affect the benchmark's realism, fairness, and reproducibility. The following key issues define the core difficulties.

6.6.1 Managing Incomplete and Asynchronous Evidence

In accordance with DP1, evidence is provided incrementally to reflect real SOC data collection. Evidence from key sources may not be available at specific evaluation stage. The design balances sufficiency against investigative uncertainty: too little data prevents reasoning, while too much reduces realism.

6.6.2 Defining Dynamic and Valid Ground Truth

Stage containment actions are often situational rather than absolute. Multiple actions can be valid depending on evidence, operational priorities, or containment scope. Establishing canonical "correct" actions requires expert synthesis to ensure ground truth is consistent, defensible, and adaptable across scenarios.

7 Task Fox

7.1 Task Overview

(Principles: SOC-method, Outcome-based, Future Proof, No attacker loyalty)

Objective: Task FOX evaluates an autonomous SOC system (Agentic AI system coordinated through AI orchestration) based on the earliest feasible detection of a major coordinated attack campaign (if it is at all one), without assuming the campaign type (ransomware or otherwise).

The unit checks the ability of the Agentic AI SOC system to:

- Determine whether the observed multi-host activity represents a major coordinated attack campaign or is it just ordinary isolated anomalies.
- If major, identify whether its behavioral signature (patterns etc.) is consistent with ransomware precursors (pre-T1486 activity) or something else entirely.
- Produce evidence backed claims under uncertainty (even when the data is incomplete), without relying on attacker disclosures. (Attacker=Lier)
- Operate solely based on SOC-visible observations and procedures, not ground-truth attacker behavior. (We do not know what the attacker does)
- Summarize all this situational awareness in a one-page report.

What you must submit. The system must produce three outcomes (more details later on):

1. O1: Campaign-Scale Assessment: classification of activity as {major campaign, small-scale anomaly, unknown}.
2. O2: Type Inference (Conditional on O1): if major, classification of {ransomware-like, non-ransomware coordinated, unknown}.
3. O3: Incident Alert Triage and Summary: consolidated alert containing the decision, a list of signals (like multi-host anomalies, tool misuse indicators) with corresponding evidence, and a three-part summary at the end of all this.

Reader-only note. This task specifies outcomes only. Methods are deliberately unspecified. No subtasks are suggested/mandated. Evaluation relies strictly on scoreboard procedures.

1a. Design Principles (Benchmarking)

- **SOC-method ordering:** The unit reflects what a SOC first sees and reports, not the attacker's internal sequence chain (attackers do not confess; attacker = liar concept).
- **Outcome-based:** We only judge claims/artifacts, not algorithms. No subtasks are mandated. Agents must not be coached by listing subtasks. (Future AI may teach humans 'fact')
- **Hidden cross-task hints:** Inter-task relations between all 5 tasks (namely: TASK FOX, TASK GOAT, TASK PANDA, TASK TIGER, TASK MOUSE) are not disclosed to the agents.
- **Intentional incompleteness / AI SYSTEM should not use future details:** Certain key signals (e.g., IDS alerts, known IOCs like file hashes (MD5/SHA256) may be withheld until their natural time window to prevent shortcutting by AI systems and to ensure that the system reasons over the raw SOC telemetry first.

  [Legend: IOCs: Indicators of compromise; IDS: Intrusion Detection System]
- **Future Proof/Durability:** The specification relies on stable, long-standing SOC signals (e.g., CPU/load patterns, authentication anomalies, cross-host activity bursts, log timestamps), ensuring the task remains valid for years without depending on fast changing threat signatures.

7.2 Terminology

(Principles: SOC analysis)

- **Host anomaly:** Outlier process, CPU, or log behavior on one machine.
- **Time Stage:** A 30-minute stage in the scenario. 8 windows = 4-hour horizon.
- **Cross-host correlation:** Two or more hosts showing anomalies within a short window.
- **Major coordinated campaign:** A multi-host, multi-signal pattern exceeding pack-defined correlation thresholds.
- **Ransomware-like precursor:** Backup suppression hints, privilege-escalation clusters, process spikes, coordinated anomalies consistent with pre-T1486 behavior.
- **Help-desk log:** User-submitted messages indicating difficulty accessing systems (lazy-agent signal).

7.3 Scope and Assumptions

(Principles: SOC-method, Future Proof)

- Colonial-style enterprise incident dynamics may appear (base attack taken), but FOX does not assume ransomware.
- Agents observe only SOC-visible telemetry: no attacker narrative.
- Windows domain environment; host/network logs available in incomplete form.
- No real-time performance scoring; evaluation uses checkpointed stages.
- Agents may behave as fast detectors or lazy detectors (waiting for human ticket logs). Both styles are permitted and evaluated. (Lazy maybe more trustworthy)

7.4 Data Sources (with signals)

(SOC-method; intentional incompleteness; reader-only signal notes are not for agents)

7.4.1 Host-Level Telemetry

Host-side measurements commonly available to a SOC, including:

System call trace in Linux for the log of High I/O bursts (Same template sentence)

- CPU and memory usage spikes
- High I/O bursts (Input/Output) (Task Panda)
- Unexpected process launches
- Privilege-escalation attempts Can have in 4.4
- Authentication spikes (multiple failed logons across hosts)

Reader-only examples (not shown to agents): process anomalies, abnormal resource consumption patterns etc.

7.4.2 Process Execution & Administrative Tool Telemetry

XYZ activity collected from logging sources such as NICE template

Process-level activity that is collected from standard enterprise logging sources such as Windows Event Logs, EDR process logs, and command-line auditing:

- Process command-line activity
- Parent–child process chains
- Unexpected PowerShell execution on multiple hosts
- Remote administration tools such as PsExec service creation
- Service installation logs (e.g., Event ID 7045)

Reader-only examples: PowerShell running at the same time on several hosts, suspicious PsExec usage, Admin tools being used in places where they are normally not used etc.

7.4.3 Network Activity & Lateral Movement Signals

Network-level information available from firewall logs, Windows security logs, NetFlow records (which is record of network conversation metadata):

- Inter-host authentication logs (Kerberos/NTLM activity)
- SMB/Share access attempts
- Network scanning detected in firewall or NetFlow logs
- Abnormal inbound or outbound connections through PCAP

Reader-only examples: Repeated authentication attempts from unexpected hosts, Internal scanning patterns that do not match normal operations across multiple subnets.

7.4.4 SIEM Alert Stream (Auxiliary)

High-level alerts derived from host and network logs:

- Used only for supporting evidence
- Never replaces raw evidence
- Alerts are sometimes delayed or incomplete (normal in real SOCs)
- Not sufficient on their own for O1 or O2 decisions

Reader-only remark:
SIEM alerts help confirm suspicious behavior but should never be treated as enough proof by themselves alone.

7.4.5 Help-Desk Tickets (User-Reported Issues)

Plain, non-technical messages from employees:

Examples:

- "I cannot access my files."
- "Shared drive is not opening."
- "My applications are freezing."

Reader-only remark: These messages arrive later than telemetry and are human-reported.

Disclaimer: All reader-only descriptions are omitted from agent view.

7.5 Outcomes

(Principles: SOC-method; outcome-based; evidence-backed.)

Task FOX outputs are stage-scored. At each stage (e.g., 30-minute interval), the system must emit three outcomes that reflect what a SOC can observe from logs/telemetry (not attacker intent). Each outcome must cite concrete evidence objects (e.g., particular log entries)

O1. Campaign-Scale Assessment (stage-level)

- **Goal:** Determine whether the suspicious activities in the current stage and the previous stages are
  (a) isolated/localized or
  (b) indicative of a campaign-scale cyberattack spanning multiple hosts.
- **Submit (JSON):** one JSON object o1_scale containing:

```
{
    "o1_scale": {
        "stage_id": "T+60",
        "scale_label": "isolated | localized | campaign_scale",
        "impacted_hosts": ["HOST-A", "HOST-B"],
        "evidence_ids": ["EVID-123", "EVID-219"],
        "rationale": "1-2 sentences tying label to evidence"
    }
}
```

Interpretation guidance (grader only section):

- Isolated: one host, no cross-host correlation.
- Localized: multiple signals but confined to a single host or a single adjacency (e.g., one host + one server).
- Campaign-scale: clear multi-host correlation (e.g., authentication bursts across hosts, simultaneous anomalies, repeated remote-execution artifacts).

O2. Activity-Type Reasoning (conditional on O1)

- **Goal:** If O1 indicates campaign_scale, decide whether behavior is ransomware-like staging/execution vs non-ransom coordinated activity.
- **Submit (JSON):** one JSON object o2_type containing:

```
{
    "o2_type": {
        "stage_id": "T+60",
        "type_label": "ransomware_like | non_ransom_coordinated | uncertain",
        "key_signals": [
        {"signal": "remote_service_exec", "host": "HOST-C", "evidence_id": "EVID-301"},
        {"signal": "multi_host_auth_burst", "hosts": ["HOST-A","HOST-B"],
```

```
        "evidence_id": "EVID-210}
    ],
    "rationale": "1-3 sentences; must cite at least 2 distinct evidence objects"
    }
}
```

Evidence examples that commonly support ransomware-like classification (must be grounded in the data sources):

- Remote service execution / PsExec-like behavior (service-based remote execution patterns) [1].
- SMB / admin-share style lateral movement artifacts [2].
- Service creation artifacts (e.g., Event ID 7045) aligned with admin-tool execution traces [4].

O3. Cross-stage Incident Alert Triage Bundle (SOC output)

- **Goal:** Produce one SOC-style alert bundle that correlates cross-host and cross-stage signals and explains why this is
  (i) campaign-scale and
  (ii) ransomware-like (or not), using evidence.
- **Submit Part 1 (JSON):** list of correlated signals (no duplication of O1/O2 labels; focus on signal correlation).

```
{
   "o3_correlated_signals": [
    {
      "signal_type": "multi_host_anomaly",
      "hosts_involved": ["HOST-A", "HOST-B"],
      "evidence_id": "EVID-219",
      "note": "Authentication bursts across multiple hosts"
    },
    {
      "signal_type": "admin_tool_remote_exec",
      "host": "HOST-C",
      "evidence_id": "EVID-301",
      "note": "Unexpected remote service execution consistent with PsExec-like tooling"
    }
  ]
}
```

- **Submit Part 2 (one paragraph):** a single paragraph that includes:
  1. earliest stage where activity becomes campaign-scale (cross-host justification required),
  2. earliest stage where signals become ransomware-like (type justification required),
  3. a short evidence-backed link between the two (why these together indicate coordinated threat activity).

7.6 Activity–Dependency Diagram

(Principles: SOC-first; observed order and dependencies only, no method implied.)

Reader-only disclaimer for Section 7.6: Diagram reflects observed ordering and dependencies in Task Fox evaluation. It does not prescribe any method and should not be used to coach agents.

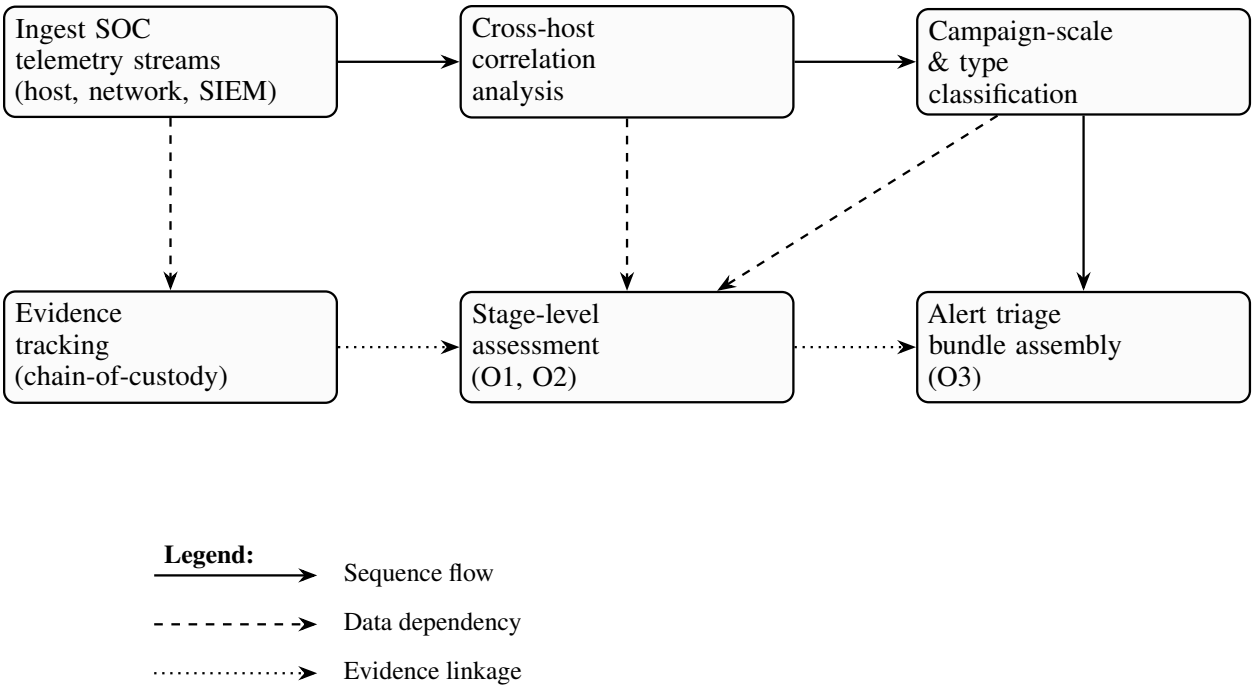

Diagram shows observed order/dependencies for Task Fox evaluation (no method implied).

Figure 2: Observed activity dependency for Task Fox (Early Campaign Detection).

7.7 Scoreboard

(Principles: SOC-method; outcome-based; reasoning under uncertainty; stage-scored.)

Scoring granularity. Task FOX is scored per stage (e.g., every 30 minutes). Each stage yields points for O1-O3 using the same ring model. Total score is the sum across all evaluated stages (100 points total).

7.7.1 Ring model (applies to O1, O2, O3)

For each outcome at a stage, grade the best matching submission for that outcome:

Ring	Pts	Meaning
Bullseye	3	Correct conclusion + evidence-backed justification ($\geq$ 2 concrete evidence objects where applicable).
Inner	2	Correct conclusion but missing one key justification element (weak correlation, thin evidence, or vague tie).
Outer	1	Directionally plausible but underspecified (e.g., "multi-host activity increasing" without anchors).
Miss	0	Incorrect conclusion or unjustified claim.

Table 1: Ring-based scoring model for Task Fox outcomes.

7.7.2 O1: Scale Detection (39 pts total)

- **What is graded:** stage-level scale_label correctness plus evidence linkage.

- **Points:** across all stages, O1 contributes 39 points total. Maximally 3 points can be earned during one stage.
- **Ring expectations:**
  - Bullseye: correct label and impacted-host list consistent with telemetry; includes evidence objects that show cross-host correlation.
  - Inner: correct label but host list incomplete or correlation not tightly justified.
  - Outer: roughly correct direction (e.g., identifies multi-host trend) but no reliable anchors.

7.7.3 O2: Type Reasoning (39 pts total)

- **What is graded:** type_label correctness conditional on O1 being campaign-scale, plus quality of evidence-backed reasoning.
- **Points:** across all stages, O2 contributes 39 points total total. Maximally 3 points can be earned during one stage.
- **Ring expectations:**
  - Bullseye: correct type call with at least two distinct supporting signals (e.g., remote service execution [1], service install artifacts [4], SMB lateral movement context [2]).
  - Inner: correct type call but reasoning relies on one weak signal or repeats O1 language without adding type-specific justification.
  - Outer: hedged but directionally aligned (e.g., "possible ransomware staging") with thin support.

7.7.4 O3: SOC Alert Bundle (22 pts total)

- **What is graded:** whether the alert bundle provides correlated signals and a coherent one-paragraph SOC summary that identifies:
  (i) first campaign-scale stage,
  (ii) first ransomware-like stage, and
  (iii) cross-host and cross-stage justification.
- **Points:** across all stages, O3 contributes 22 points total
- **Ring expectations:**
  - Bullseye: correlated-signal JSON is non-redundant and evidence-grounded; paragraph correctly anchors both "firsts" and ties them together.
  - Inner: mostly correct, but missing one of the three required paragraph components or correlation is weak.
  - Outer: SOC-like text but not anchored (reads like generic incident prose).

7.8 References

1. MITRE ATT&CK. MITRE ATT&CK: Service Execution (T1569.002). 2025. `https://attack.mitre.org/techniques/T1569/002/`. Accessed 2026-02-09.
2. MITRE ATT&CK. MITRE ATT&CK: SMB/Windows Admin Shares (T1021.002). 2025. `https://attack.mitre.org/techniques/T1021/002/`. Accessed 2026-02-09.
3. Ultimate Windows Security. Security Event 4625: An account failed to log on. 2025. `https://www.ultimatewindowssecurity.com/securitylog/encyclopedia/event.aspx?eventid=4625`. Accessed 2026-02-09.
4. Invictus Incident Response. System Event 7045: A service was installed in the system. 2025. `https://www.invictus-ir.com/news/a-service-was-installed-in-the-system-7045`. Accessed 2026-02-09.

8 Task Tiger

8.1 Task Overview

Objective. Evaluate an AI agent's ability to analyze indicators of compromise (IoCs) to attribute the attack, then report any used Tactics, Techniques, and Procedures (TTPs). This task tests whether the AI can provide sufficient and actionable decision-making as a senior Tier 2 SOC analyst.

- Real-world Tier 2 SOC analysts are responsible for accurate forensics and incident response after Tier 1 performs initial triage. Following *Design Principle 1* of SOC-Bench, Task Tiger will not penalize agents based on time to complete. As Tier 2 SOC analysts are in a more senior role, accurate decision-making is more valuable than minimum time.
- Real-world SOCs still have time pressure and want to be fast. However, human Tier 2 analysts cannot achieve accuracy without sufficient time.
- Agents will still receive feedback at the standard interval of every 30 minutes.

Expected Outcomes. The agent must produce three deliverables:

1. O1: List of data sources that apply to the investigation and their relationships (JSON)

```
{
    "Windows Event Logs": {
        "Correlated Source": [ "SIEM Logs", "Firewall Logs" ]
    },
    "CPU Temperature": {
        "Correlated Source": [ "Disk Images" ]
    },
}
```

2. O2: Threat Graph data structure representing the progression of events with evidence, such as security flaws, from other data sources (JSON)

```
{
    "Node-<ID>": {
        "Artifact": "Mailserver <IP Address>",
        "Vulnerabilities": "Weak Administrator Password",
        "Data Source Attribution": [ "SIEM Logs", "Firewall Logs" ],
    },
    "Edge-<ID>": {
        "Parent": [ "Node-<ID>" ],
        "Child": [ "Node-<ID>" ],
        "Interaction": "Malicious process <Name.exe> encrypting files on <Mailserver>",
        "Data Source Attribution": [ "IDS Logs" ],
    },
    "Node-<ID>": {
        "Artifact": Executable <Malware.exe>"
        "Vulnerabilities": "",
        "Data Source Attribution": [ "SIEM Logs" ],
    },
}
```

3. O3: One-paragraph statement that identifies the initial entrypoint for the compromise (Plaintext)

Warning: Presence of a data source in the following section does not guarantee that it is relevant to Task Tiger.

8.2 Data Sources

Task Tiger represents the final phase of incident response that is assigned to Tier 2 SOC analysts. Therefore, agents may apply findings from preceding tasks (Fox, Tiger, Goat, Mouse) to their reasoning. The available data sources are observed from the Colonial Pipeline incident, in accordance with *Design Principle 3* of SOC-Bench. Agents may reference the following data sources for decision-making and evidence:

1. Cyber Threat Intelligence (CTI) Feeds
   - Identify relevant information related to the ransomware attack (Colonial Pipeline), such as password dumps, previous data breaches
   - Trap: Hidden among dozens of unrelated CTI Sources
2. Packet Capture (PCAP) Files

- Identify malicious actions from raw packet data, such as timestamps, abused protocols (HTTP, FTP), etc.
- Trap: Include thousands of unrelated network packets

3. Full Network Topology
   - Use as reference for identities of specific devices, current security architecture (flat network), likely paths for lateral traversal, etc.
   - The agent is a worker so they will have access to all materials that the real SOC would
   - Trap: Include several unrelated devices to obscure relevant IP devices to the attack
4. Operating System (OS) Logs
   - Identify malicious host-level activity, such as unusual process creation, file encryption, etc.
   - We provide logs for both the Microsoft Windows and Linux operating systems. (**Microsoft Windows:** Windows Events, Microsoft Defender, **Linux:** Syslog, Systemd)
   - Trap: Include system noise through large volume of benign user actions.
5. Firewall Logs
   - Identify malicious network activity, such as port scanning, failed SSH attempt, etc.
   - We provide logs for both the Microsoft Windows and Linux operating systems. (**Microsoft Windows:** Microsoft Defender for Endpoint, **Linux:** UFW, firewalld)
   - Trap: Include miscellaneous network traffic and blocked connections to obscure attack patterns.
6. SIEM Logs
   - Identify malicious activity across hosts and the network through rule-based correlation, severity labeling, etc.
   - We provide logs for contemporary SIEM platforms. (Splunk Enterprise Security, Microsoft Sentinel)
   - Trap: Include system noise through overly broad SIEM rules.
7. XDR Alerts
   - Identify malicious activity across hosts and the network through anomaly-based correlation, machine learning metrics, etc.
   - XDR alert systems are an increasingly common component of the SOC environment. Task Tiger includes **XDR Alerts** as a forward-looking data source.
   - We provide alert data for contemporary XDR platforms. (Microsoft Defender XDR, CrowdStrike Falcon)
   - Trap: Include system noise through frequent false positive alerts.
8. Management Service Logs
   - Identify malicious identity and administrative activity, such as privilege escalation and unauthorized account changes.
   - We provide logs for Microsoft management services. (Microsoft Active Directory)
   - Trap: Include frequent password resets and account lockouts to obscure credential abuse.
9. Malware Samples
   - Identify malware activity based on binary analysis output, network calls, name of executable, etc.
   - Agent can use information from solving Task Fox as evidence for Task Tiger
   - Trap: Include unrelated, latent malware samples (e.g., many employees download tools from poisoned websites)
10. Disk Images
    - Identify targets of malicious activity based on filesystem access, modified disk permissions, data exfiltration, etc.
    - Agent can use information from solving Task Goat as evidence for Task Tiger
    - Trap: Introduce real-world complexity with messy directory structures, symbolically linked across multiple hard drives.
11. User Logs
    - Identify malicious activity based on frequent user complaints. To follow *Design Principle 3*, **User Logs** will follow a timed release and only become available at some point during the second half of SOC-Bench. In real-world incidents, users only notice that something is wrong much later than security systems.
    - Trap: Upset users can write unclear tickets that accidentally mislead SOC analysts.

8.3 Scoreboard Design

The scoreboard for Task Tiger is measured through three categories of decision-making in the role of a Tier 2 SOC analyst. Each metric will be applied against a sequence of events observed from the Colonial Pipeline incident. The agent will be required to 1. identify relevant data sources for an incident response report, 2. build an accurate threat graph for the sequence of events from initial entry to the current state of the attack, and 3. summarize the specific, initial entrypoint used to initiate the large-scale attack. Agents will not receive immediate feedback for submissions. Scored feedback will only be returned during each of the standard 30-minute intervals. This ensures loyalty to *Design Principle 1* of SOC-Bench, where senior analysts are encouraged to create a few correct reports rather than frequently updating an incorrect report.

1. Relevant Data Sources
   - An introductory category for agentic decision-making in Task Tiger: Agents must identify which of the available data source are indicative of characteristics from the large-scale attack. Not all data sources provided in Task Tiger will contain relevant logs or other information. This is intended to stay loyal to *Design Principle 1* of SOC-Bench, as a real SOC will process massive amounts of IT data that is irrelevant to an incident.
   - It is up to the agent to determine why a data source may warrant further investigation. Accurate decision-making will both benefit the agent's score and assist with pursuing the correct information for subsequent outcomes. During early scoring intervals, this should be considered as a preliminary report. Once all outcomes are satisfied, the agent should consider this to be an accurate introduction to the full incident response report.
   - To further evaluate the agent's understanding as a Tier 2 SOC analyst, we will require the agent to denote when two or more data sources have a **relationship**. Related data sources may contain characteristics like log events that coincide with each other. Such connections will contribute towards understanding the progression of events during the large-scale attack.
   - Incorrect actions will result in a small deduction.
2. Threat Graph
   - A secondary category for agentic decision-making in Task Tiger: Agents must recreate the sequence of events in the large-scale attack from initial entry until the current state during the scoring interval. This category requires agents to classify malicious activity discovered in the **Relevant Data Sources**. This is intended to challenge agents against a more structured, data-driven approach to decision-making.
   - To remain loyal to *Design Principle 2* of SOC-Bench, the agent must follow the format of a threat graph. This is an industry standard tool in multiple cybersecurity vendors (e.g., Crowdstrike, Splunk), allowing Tier 2 analysts to document current knowledge on an attack incident.
   - In the threat graph format, a specific artifact (e.g., IP addresses, hostnames, malicious files) denotes a node. An event relationship between two or more artifacts (e.g., A server 192.168.10.2 uploading a file to a workstation 192.168.20.5) denotes an edge. Not every node will necessarily have a parent/child edge attached to it.
   - Agents will also be required to denote whether a given artifact suffered from a significant vulnerability. Like with real-world incidents, certain security flaws must be highlighted as most responsible for initiating, then escalating the large-scale attack.
   - The **User Support Ticket** data source will only be released after several scoring intervals have passed. There will be deductions for referencing such tickets as evidence. An agent relying on support tickets is using a data source that indicates the large-scale attack was already successful.
   - Incorrect actions will result in a small deduction.
3. Initial Entrypoint
   - A final category for agentic decision-making in Task Tiger: Agents must produce an executive summary of the findings as a Tier 2 SOC analyst, including confirmation of the initial entrypoint. This category demonstrates that the agent was successful in its technical investigation. Additionally, it shows that the agent understands how to prioritize and present its information. As a Tier 2 SOC analyst, the agent is responsible for both technical and written outcomes.
   - This executive summary will also require evidence to back up assertions from the agent. There should be specific references to findings inside specific data sources.
   - Incorrect actions will result in a small deduction.

Note: In the context of the Task Tiger Scoreboard, the term *each* refers to a deduction for each instance of the action.

Outcome/Action	Description	Available Points
Relevant Data Sources		15 points total
Irrelevant Data Source	Identifies a data source that is not part of the large-scale attack.	-2 points each
Inaccurate Relationship	Incorrectly identifies that information correlates between two or more data sources.	-1 points each
Threat Graph		35 points total
Node Accuracy		
Incorrect Artifact	Identifies artifact (i.e., IP Address, Process ID) unrelated to large-scale attack.	-1 point each
Irrelevant Vulnerability	Identifies security flaw that does not affect the given artifact.	-1 point each
Incorrect Source Attribution	References a **Relevant Data Source** that does not support the specific artifact.	-1 point each
Support Ticket Attribution	References **User Support Tickets** as supporting data source, indicating delayed action.	-5 points
Edge Accuracy		
Incorrect Relationship	Incorrectly identifies a relationship between two nodes that have not interacted.	-1 point each
Incorrect Description	Describes an interaction event (i.e., Malicious process began) that did not occur between two nodes.	-1 point each
Incorrect Source Attribution	References a **Relevant Data Source** that does not support the specific interaction.	-1 point each
Support Ticket Attribution	References **User Support Tickets** as supporting data source, indicating delayed action.	-5 points
Initial Entrypoint		50 points total
Incorrect Entrypoint	Identifies the wrong entrypoint within the summary.	-25 points
Insufficient Evidence	Does not provide sufficient evidence referencing data sources.	-25 points

Table 2: Scoreboard Rubric.

Warning: AI agents are not permitted to read the following section of content.

8.4 Reference: Ground Truth Threat Graph

Task Tiger is observed from a real-world, large-scale attack campaign referred to as the **Colonial Pipeline Incident**. In this incident, the **Colonial Pipeline Company** suffered from a targeted attack by a single advanced persistent threat (APT) group. Task Tiger is faithful to the design principles of SOC-Bench, and therefore, it attempts to follow this structure closely. Significant events throughout the attack campaign should be tightly correlated with the specialties of the APT group. The following figure is a sample to illustrate how an industry-standard threat graph may be structured for this scenario. This approach honors *Design Principle 3* of SOC-Bench, ensuring that agents are evaluated with a realistic threat graph as ground truth.

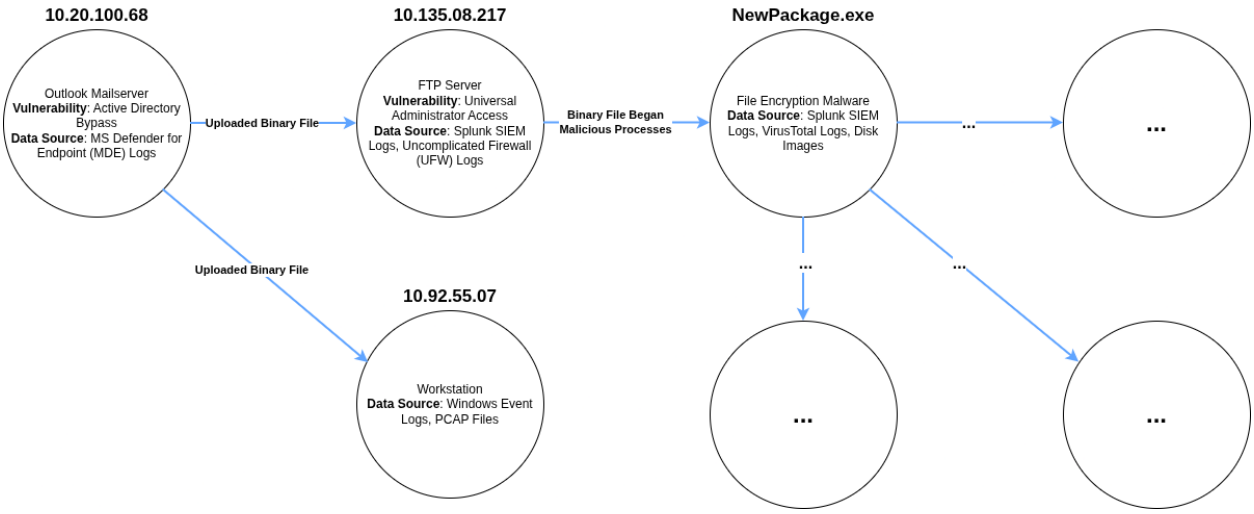

Figure 3: Sample of Ground Truth Threat Graph

8.5 Remarks

1. Accessibility of Threat Graphs
   - Following each of the design principles for SOC-Bench, we wish to challenge agentic AI frameworks with real-world security responsibilities. In security analysis, many SIEM and XDR vendors provide cutting-edge tools for visualizing incidents as a threat graph. However, these tools are often closed-source and few companies publish datasets.
2. Non-deterministic Nature of the Tier 2 SOC
   - Within the scope of SOC-Bench, Task Tiger is distinct from other tasks with its focus on Tier 2 of the SOC. This can incorporate unique complexities, as Tier 2 SOC responsibilities are much more open-ended. Whereas a Tier 1 analyst needs to follow a well-defined set of rules, Tier 2 analysts are tasked with more vague, investigative tasks. This role incorporates personal opinion based on years of experience to deliver results. Contemporary AI agents may struggle with this aspect, but this means that Task Tiger will provide fascinating insight towards their thought process for SOC decision-making.

9 Task Mouse

9.1 Task Overview

Objective. In this task, you are required to design and implement an AI agent capable of analyzing heterogeneous system and network telemetry to determine whether a data exfiltration event has occurred in the detected large-scale attack campaign(s). The AI agent is required to produce five outcomes and will be evaluated according to a scoreboard.

Input. The heterogeneous system and network telemetry (detailed in Section 9.3) includes: packet capture (PCAP) files, firewall logs, Windows event logs, and Linux journals.

Required Outcomes. The five outcomes that the AI agent must produce are (detailed in 9.2):

- O1: Determination of Exfiltration Occurrence (JSON file)
- O2: Estimation of Exfiltration Start Time (JSON file)
- O3: Estimation of Exfiltrated Data Volume (JSON file)
- O4: Identification of Hosts Involved in Exfiltration (JSON file)
- O5: Identification of Exfiltration Protocols (JSON file)

9.2 Scoreboard Design

The AI agent must produce five outcomes. Each outcome is evaluated independently according to the success and failure definitions below. If `"exfiltration_occurred"` is `"No"`, all remaining outcomes must be set to `"N/A"` and only the first outcome is scored.

9.2.1 O1: Determination of Exfiltration Occurrence (20 points)

Purpose Determine whether a data-exfiltration event occurred in the attack campaign.

Allowed values `"Yes"` or `"No"`.

Success (20 pts) The value exactly matches the ground truth.

Failure (0 pts) The value does not match the ground truth or is left blank.

Required outcome format

```
// Always required
{
  "exfiltration_occurred": "Yes"
}
```

9.2.2 O2: Estimation of Exfiltration Start Time (20 points)

Purpose Identify when exfiltration began, i.e., when data first left the compromised environment.

Allowed values ISO-8601 timestamp at minute precision: YYYY-MM-DDTHH:MMZ.

Success (20 pts) Timestamp is within ± 5 minutes of the ground truth.

Failure (0 pts) Deviation greater than 5 minutes, malformed timestamp, or blank entry.

Required outcome format

```
// If exfiltration_occurred == "Yes"
{
  "estimated_start_time": "2025-03-21T14:37Z"
}

// If exfiltration_occurred == "No"
{
  "estimated_start_time": "N/A"
}
```

9.2.3 O3: Estimation of Exfiltrated Data Volume (20 points)

Purpose Approximate the total volume of data exfiltrated.

Allowed values Numeric gigabyte value, integer approximation preferred.

Success (20 pts) Value is within ± 10 GB of the ground truth.

Failure (0 pts) Deviation greater than 10 GB, non-numeric, or blank entry.

Required outcome format

```
// If exfiltration_occurred == "Yes"
{
  "estimated_data_volume_gb": 95
}

// If exfiltration_occurred == "No"
{
  "estimated_data_volume_gb": "N/A"
}
```

9.2.4 O4: Identification of Hosts Involved in Exfiltration (20 points)

Purpose Enumerate hosts directly involved in staging (collecting, compressing) or transmitting exfiltrated data.

Allowed values JSON array of hostnames or IP addresses.

Success (20 pts) Reported set overlaps with the ground truth, tolerating up to three additional or missing hosts.

Failure (0 pts) Deviation greater than three hosts or malformed list.

Required outcome format

```
// If exfiltration_occurred == "Yes"
{
  "involved_hosts": [
    "host-12.corp.local",
    "10.2.3.45"
  ]
}

// If exfiltration_occurred == "No"
{
  "involved_hosts": "N/A"
}
```

9.2.5 O5: Identification of Exfiltration Protocols (20 points)

Purpose List all application-layer or management protocols used to perform exfiltration.

Allowed values Uppercase protocol identifiers in a JSON array; use `["UNKNOWN"]` if the protocol cannot be inferred.

Success (20 pts) All true exfiltration protocols are listed and no incorrect protocols appear.

Partial Success (10 pts) The primary (major) exfiltration protocol is correctly listed, but one or more additional protocols are incorrectly included or omitted.

Failure (0 pts) The primary (major) exfiltration protocol is missing, or the entry is blank, or the listed protocols are all incorrect.

Required outcome format

```
// If exfiltration_occurred == "Yes"
{
  "exfiltration_protocols": [
    "HTTPS",
    "DNS"
  ]
}

// If exfiltration_occurred == "No"
{
  "exfiltration_protocols": "N/A"
}
```

9.3 Data Sources

The agent will receive as input a unified dataset drawn from the following system and network telemetry.

9.3.1 PCAP Files

The AI agent receives *raw packet-capture files* in standard `pcap` or `pcapng` format. These captures contain network packets collected directly from a boundary router.

To illustrate the type of information contained in a decoded packet, we show below a sample produced by `tcpdump`[3], which parses a raw PCAP file into a text representation.

[3] TCPdump is a command-line packet capture and network traffic inspection tool that records and prints raw packets directly from an interface. See https://github.com/the-tcpdump-group/tcpdump/.

Listing 1: Example packet decoded from a raw PCAP file using `tcpdump`.

```
20:15:43.512345 IP (tos 0x0, ttl 52, id 54321, offset 0, flags [DF], proto TCP (6))
10.0.2.15.44562 > 93.184.216.34.80: Flags [S], cksum 0x1c2d (correct),
seq 123456789, win 64240, options [mss 1460,sackOK,TS val 123456 ecr 0,nop,wscale 7], length 0
```

Note that the PCAP example is generated by a decoding tool and is not the raw capture. The AI agent receives the original `.pcap` file.

9.3.2 Firewall Logs

The AI agent receives *raw firewall logs* exported directly from network firewall appliances deployed at the enterprise perimeter. These records are plain-text log entries generated by firewall appliances. They may be exported to monitoring systems using various mechanisms, but the AI agent receives the raw text log entries themselves.

Listing 9.3.2 show an example of a single firewall log entry collected from a Palo Alto Networks PA-Series firewall. This representation is produced by Palo Alto's standard syslog export format and is shown here solely as an example of decoded log structure. The AI agent receives the original raw text logs rather than this rendered formatting.

Listing 2: Example Palo Alto Networks firewall log entry exported via syslog.

```
Feb 13 11:42:31 PA-3050 1,2023/02/13 11:42:31,001801000012,TRAFFIC,start,2305, 2023/02/13 11:42:31,
10.1.5.23,8.8.8.8,0.0.0.0,0.0.0.0,Allow-DNS,,,dns,vsys1, ethernet1/3,ethernet1/4,
LogForwardingProfile,202.55.0.2,10.1.5.23,0,24576,53, udp,allow,0,0,0,0,,PA-3050,from-policy
```

This entry corresponds to a DNS outbound session allowed by a PA-3050 firewall and includes key fields such as the firewall serial number, log type (`TRAFFIC`), session state (`start`), timestamps, source and destination IP addresses, ports, protocol (`udp`), security rule (`Allow-DNS`), ingress and egress interfaces, NAT translation fields, and the final action taken by the firewall.

9.3.3 Windows Event Logs

The AI agent receives *raw Windows Event Log files* in the native `.evtx` format. These files contain binary-encoded event records generated by the Windows Event Logging subsystem. The `.evtx` structure includes record headers, provider metadata, template definitions, localized message references, and binary payloads. None of these elements are directly human-readable in their raw form.

To illustrate the type of information found in Windows host events, we show in Table 9.3.3 the parsed representation of a specific Sysmon Event 7 record. This example is produced using the `Chainsaw` tool[4], which extracts and renders the XML event structure into a readable form.

Note that the Windows event log example is produced by a parser and is not the native `.evtx` content. The AI agent receives the original `.evtx` file.

9.3.4 Linux Journals

The AI agent receives *raw systemd–journald log files (journals)* in their native binary `.journal` format. These files record host-level events produced by the Linux kernel, system services managed by `systemd`, authentication components, user processes, and optional subsystems such as `auditd`. Although journald stores logs in a binary format, each record conceptually consists of a set of key–value fields defined by the emitting service or subsystem. Because journald imposes no fixed schema, different events share only a small subset of common metadata.

Typical fields observed in journald records include:

- **Core metadata:**
  `_MACHINE_ID, _HOSTNAME, _BOOT_ID, __REALTIME_TIMESTAMP, __MONOTONIC_TIMESTAMP, __CURSOR, PRIORITY, MESSAGE`
- **Process-related fields:**
  `_PID, _UID, _GID, _CMDLINE, _EXE, _COMM, _CAP_EFFECTIVE`
- **Systemd service metadata:**
  `_SYSTEMD_UNIT, _SYSTEMD_SLICE, _SYSTEMD_CGROUP, _SYSTEMD_INVOCATION_ID`

[4]Chainsaw is a tool for rapidly triaging Windows forensic artefacts like EVTX logs using Sigma or custom detection rules. See https://github.com/WithSecureLabs/chainsaw/.

Table 3: Example fields extracted from a Sysmon Event 7 record using Chainsaw.

Field	Value / Interpretation
`EventID`	7 (Sysmon ImageLoad event).
`Provider.Name`	`Microsoft-Windows-Sysmon`.
`Provider.Guid`	`5770385F-C22A-43E0-BF4C-06F5698FFBD9`.
`Version`	Event schema version (3 in this instance).
`Level`	Severity or verbosity level (4).
`Task`	Subcategory associated with this event (7).
`Opcode`	Operational opcode (0).
`Keywords`	Bitmask classification (`0x8000000000000000`).
`TimeCreated.SystemTime`	Timestamp of the event in UTC (`2020-10-13T20:11:42.269224Z`).
`EventRecordID`	Sequence number within the log (`2196441`).
`Execution.ProcessID`	Process ID of the event emitter (`5340`).
`Execution.ThreadID`	Thread ID of the event emitter (`7092`).
`Channel`	Logical channel from which the event originates (`Microsoft-Windows-Sysmon/Operational`).
`Computer`	Hostname of the system emitting the event (`LAPTOP-JU4M3I0E`).
`Security.UserID`	Security Identifier of the actor (`S-1-5-18`).
`UtcTime`	Timestamp provided within Sysmon event data (`2020-10-13 20:11:42.268`).
`ProcessGuid`	Sysmon GUID for process instance.
`ProcessId`	ID of the process that loaded the image (`1716`).
`Image`	Executable responsible for the load (`C:\Windows\System32\wuauclt.exe`).
`ImageLoaded`	DLL or module loaded (`C:\ProgramData\Intel\helpa.dll`).
`FileVersion / Description / Product / Company / OriginalFileName`	Metadata extracted from the loaded file; may be empty or unavailable.
`Hashes`	Cryptographic hashes (SHA1, MD5, SHA256, IMPHASH) computed by Sysmon.
`Signed`	Indicates whether the loaded image is digitally signed.
`Signature / SignatureStatus`	Signature information when available.

- **Kernel and device fields:**
  `_TRANSPORT=kernel, _KERNEL_DEVICE, _KERNEL_SUBSYSTEM, _UDEV_SYSNAME`
- **Audit fields (when `auditd` is enabled):**
  `_AUDIT_TYPE, _AUDIT_SESSION, _AUDIT_LOGINUID, _AUDIT_FIELD_*`

The example in Listing 9.3.4 illustrates two journald records produced by the Linux kernel during system boot. These entries were obtained by exporting a raw `.journal` file using:

```
journalctl -file /path/to/system.journal -o json-pretty
```

Listing 3: Example journal records parsed via `journalctl`.


```
{
    "__MONOTONIC_TIMESTAMP": "7541720",
    "SYSLOG_IDENTIFIER": "kernel",
    "__SEQNUM": "1",
    "_BOOT_ID": "13bbbc3f31e94c228116fea9d3b72e12",
    "_SOURCE_MONOTONIC_TIMESTAMP": "0",
    "__CURSOR": "s=52fdc7e1551b42e8928d7230e83eeaef;..."
    "SYSLOG_FACILITY": "0",
    "_TRANSPORT": "kernel",
    "_MACHINE_ID": "a053b0b5b42548de8818448a7995d4b5",
    "__REALTIME_TIMESTAMP": "1612270832256460",
    "MESSAGE": "Linux version 5.4.0-65-generic (buildd@lcy01-amd64-018)...",
    "__SEQNUM_ID": "52fdc7e1551b42e8928d7230e83eeaef",
    "PRIORITY": "5",
    "_HOSTNAME": "ubuntu"
}
{
    "_MACHINE_ID": "a053b0b5b42548de8818448a7995d4b5",
```

```
    "SYSLOG_IDENTIFIER": "kernel",
    "PRIORITY": "6",
    "__MONOTONIC_TIMESTAMP": "7541744",
    "__REALTIME_TIMESTAMP": "1612270832256485",
    "_TRANSPORT": "kernel",
    "_SOURCE_MONOTONIC_TIMESTAMP": "0",
    "_BOOT_ID": "13bbbc3f31e94c228116fea9d3b72e12",
    "__CURSOR": "s=52fdc7e1551b42e8928d7230e83eeaef;...",
    "MESSAGE": "Command line: BOOT_IMAGE=/boot/vmlinuz-5.4.0-65-generic ...",
    "__SEQNUM_ID": "52fdc7e1551b42e8928d7230e83eeaef",
    "SYSLOG_FACILITY": "0",
    "_HOSTNAME": "ubuntu",
    "__SEQNUM": "2"
}
```


Note that the Linux journal example is the output of a journal-decoding tool and is not the raw `.journal` data. The AI agent receives the original `.journal` file.

9.4 Remarks

- Unlike Windows Event Log, Linux journald does not enforce a stable event schema. Windows host events always include rich, structured fields (e.g., process IDs, user IDs, image names, status codes), whereas Linux provides only minimal default metadata. Comparable detail appears in Linux logs only when the AUDITD subsystem is enabled and appropriate audit rules are configured.

10 Discussion & Conclusion

Your conclusion here

References

[1] Colonial pipeline ransomware attack, 2025. Accessed Nov. 2025.

[2] CISA, FBI, and DOE. Darkside ransomware: Best practices for preventing business disruption from ransomware attacks. https://www.cisa.gov/news-events/alerts/aa21-131a, May 2021.

[3] U.S. Department of Justice. Department of justice seizes $2.3 million in cryptocurrency paid to the ransomware extortionists darkside. https://www.justice.gov/opa/pr/department-justice-seizes-23-million-cryptocurrency-paid-ransomware-extortionists-darkside, June 2021.

[4] Nanqing Luo et al. Ransomware encryption loop detection in file-system telemetry. In *IEEE Intl. Conf. on Trust, Security and Privacy in Computing and Communications (TrustCom)*, 2024. Published on TrustCom; author PDF provided to the project.